Cloud Based File Security System with Hybrid Encryption

PPD-II Second Review | PID: 23CSEP03

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Outline

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2. Implementation Progress
3. Technical Approach
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System / Design Development

System Overview — CloudVault

- A cloud file storage system with hybrid encryption (AES + RSA)
- Files are encrypted on the server during upload
- Files are decrypted **in the browser** during download
- User's RSA private key never leaves the browser
- **Zero-effort sharing**: server re-wraps the AES key for the recipient

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System Architecture

Three-Tier Design

Frontend	React + Vite, Web Crypto API, Axios
Backend	Spring Boot 3 (Java 17), Spring Security, JPA
Storage	H2 Database + Encrypted Files on Disk

Key Design Decisions

- **Key Envelope Table**: per-file, per-user encrypted AES key entries (OWNER / SERVER / SHARED)
- Server holds its **own RSA key pair** to enable sharing without involving the owner
- Encrypted AES key and IV sent via **custom HTTP headers** during download

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Database Design

Three Core Entities

- **User**: username, email, password hash (BCrypt), RSA public key (Base64)
- **EncryptedFile**: original filename, stored filename (UUID), file size, content type, IV, owner ID
- **FileKeyEntry**: file ID, user ID, encrypted AES key, key type (OWNER / SERVER / SHARED)

Access Control

Each file has per-user encrypted AES key entries. Adding a user = creating a new key entry. Revoking access = deleting their key entry.

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Implementation Progress

What Has Been Completed

Backend (Spring Boot)

- **User Authentication**: Registration with RSA public key, login with JWT, BCrypt password hashing
- **Encryption Engine**: AES-256-GCM file encryption, RSA-2048-OAEP key wrapping using Java Cryptography Architecture
- **File Upload & Storage**: Encrypted upload, UUID-based storage on disk, Key Envelope Table (OWNER + SERVER entries)
- **File Download**: Returns encrypted file + encrypted AES key + IV via custom HTTP headers
- **File Deletion**: Owner-only deletion with cleanup of disk file, key entries, and DB record

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What Has Been Completed (cont'd)

Frontend (React + Vite)

- **RSA Key Generation**: Web Crypto API generates RSA-2048 key pair in browser; private key stored in IndexedDB
- **Login & Register Pages**: Dark-mode glassmorphism UI with form validation
- **Dashboard**: Tabbed file manager (My Files / Shared With Me) with stat cards
- **File Upload**: Drag-and-drop with circular progress indicator and 50MB size check
- **Client-Side Decryption**: Full pipeline — RSA decrypt AES key $\rightarrow$ AES decrypt file $\rightarrow$ browser download

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Next Steps — File Sharing

Sharing Module (To Be Implemented)

- **Share File**: Server decrypts AES key using its RSA private key, re-encrypts with recipient's RSA public key, stores SHARED key entry
- **Revoke Access**: Delete recipient's key entry from the database
- **Share Modal UI**: Enter recipient username, view shared users list, one-click revoke

Why Sharing is the Final Step

The encryption engine, key management, and file upload/download are already complete. Sharing simply **re-uses** the existing key wrapping infrastructure to create a new key entry for the recipient.

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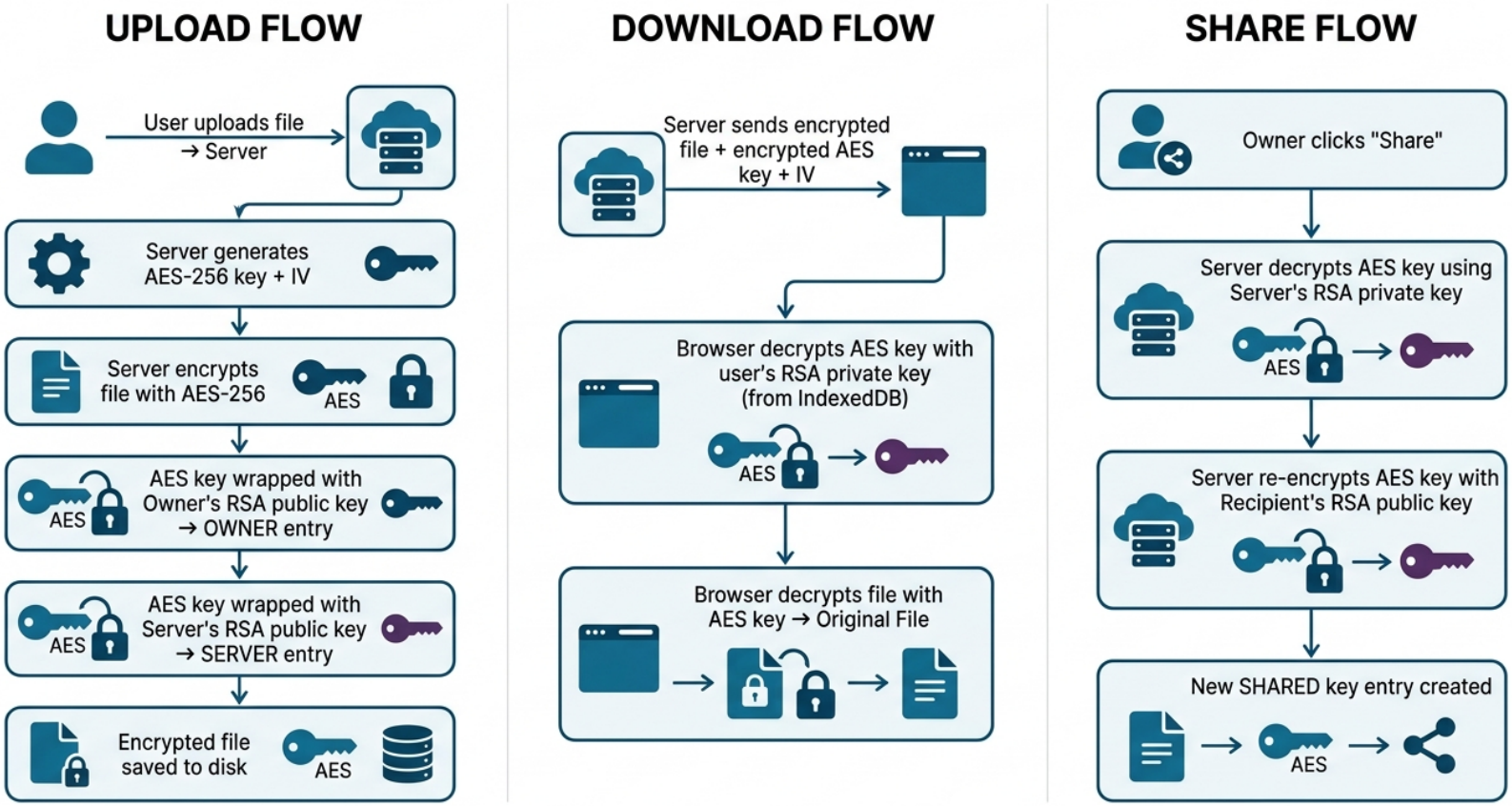
Technical Approach

Why Hybrid Encryption?

- **AES alone**: Fast for large files, but key distribution is insecure
- **RSA alone**: Secure key exchange, but too slow for large files
- **Hybrid**: AES-256 encrypts the file (fast) + RSA-2048 wraps the AES key (secure)

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Encryption Workflow



Security Layers

Multi-Layer Security

Passwords	BCrypt hashing with random salt
Sessions	Stateless JWT tokens (24h expiry)
API Protection	Spring Security filter + JWT validation
Files at Rest	AES-256-GCM encrypted on disk
Key Protection	RSA-wrapped AES keys per user
Client Keys	RSA key pair stored in browser IndexedDB
Server Keys	RSA key pair stored as files on disk

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Preliminary Results

Functional Verification

Test Results (Completed Features)

Test Case	Status	Observation
User Registration (RSA keygen)	✓	Key pair generated in browser
User Login (JWT flow)	✓	Token stored, API access works
File Upload (AES encryption)	✓	Encrypted file stored on disk
File Download (client decrypt)	✓	Original file recovered
File Deletion (owner only)	✓	Disk + DB cleaned up
Unauthorized access	✓	Returns 401, redirects to login

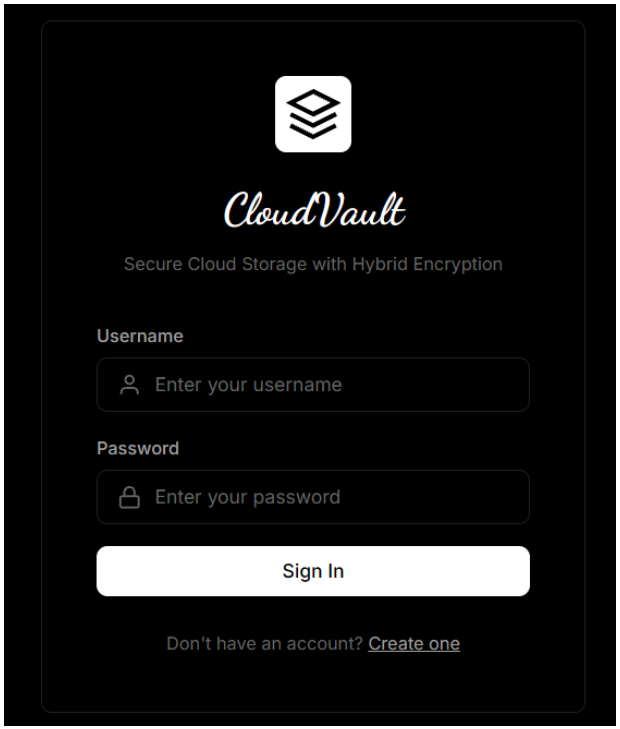
Pending Tests

File sharing and access revocation tests will be conducted after the sharing module is implemented.

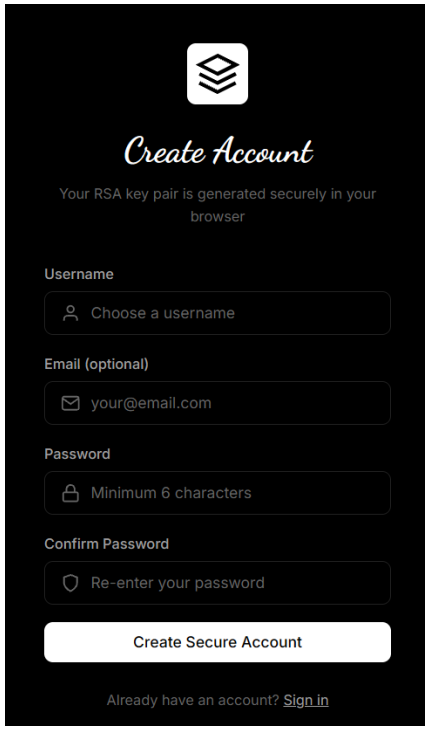
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Demo — Login & Registration

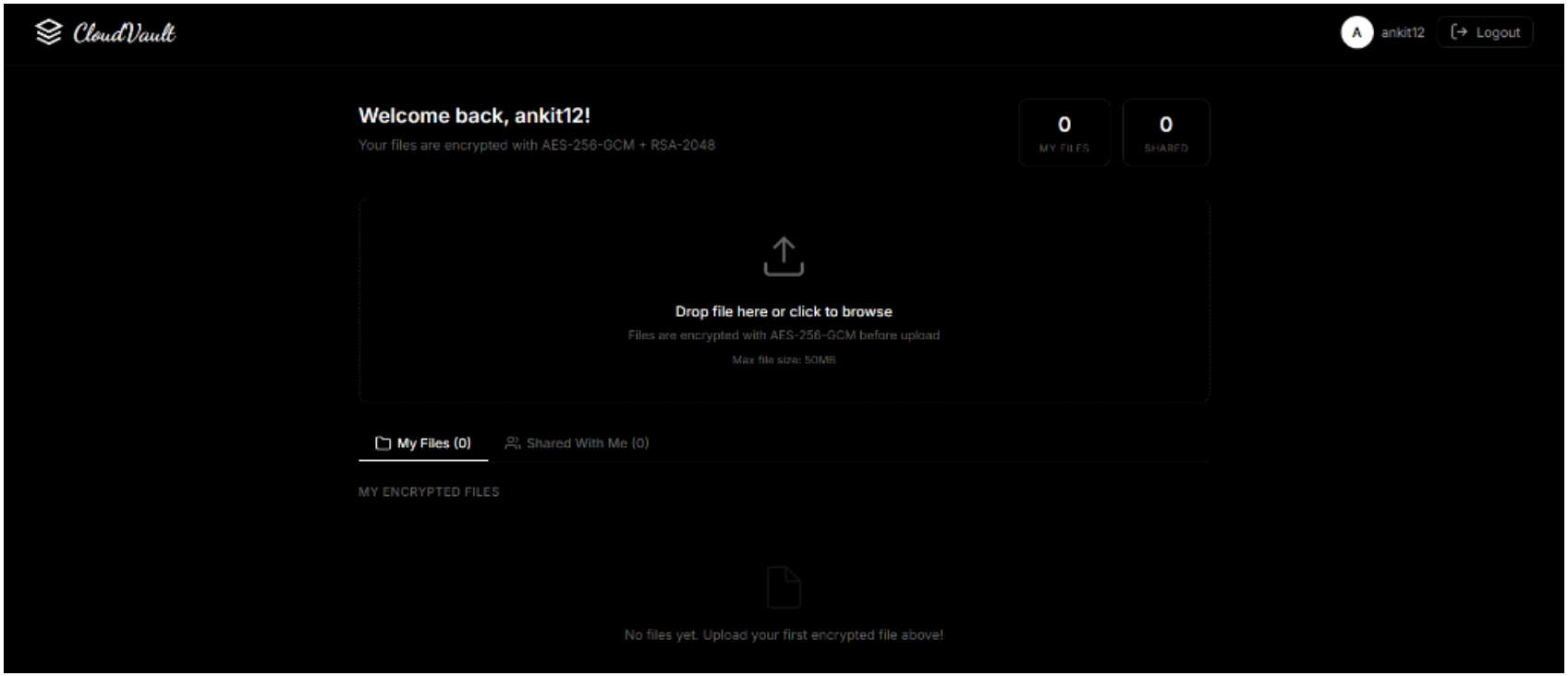
Login Page



Register Page



Demo — Dashboard



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Thank You

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